Abhay Pratap Singh

abhaypratapsingh1006@gmail.com +91 7011984714 LinkedIn GitHub Portfolio

Education

Ajay Kumar Garg Engineering College, Ghaziabad Ghaziabad, India
Bachelor of Technology in Computer Science and Engineering (AI and ML) *2021 - 2025*

Experience

Software Development Engineer in Test (SDET) Bangalore, India
CodeChef *January 2025 – Present*

- **Course & Curriculum Development**: Designed and authored comprehensive courses on Node.js, Express.js, Python Flask, and SQL, implementing end-to-end project-based learning modules.
- **DSA Engineering & Content Strategy**: Engineered a new DSA learning path and developed 1000+ high-quality problems. Produced animated visual tutorials using **Manim** and AI-driven voiceovers via **11Labs** to enhance learner retention.
- **Product Management**: Managed product execution by aligning stakeholder requirements, prioritizing deliverables, and coordinating cross-functional teams to deliver educational software and content on schedule.
- **Contest Management (Monday Munch)**: Spearheaded the 'Monday Munch' contest series making it rated, scaling participation to over 5,000 active contestants and 10,000+ unique monthly participants.

Projects

SignLingo (*Aug 2024 – May 2025*)

- **Leadership & Vision**: Led a 4-member team to build a real-time Indian Sign Language translator bridging gaps for the deaf-mute community.
- **ML-Powered Translation**: Trained a custom **MobileNetV2** CNN using **PyTorch**, achieving **90%+ accuracy** in real-time gesture recognition via webcam.
- **Research Publication**: Published *"Breaking Barriers: SignLingo as a Two-Way Communication Aid"* detailing the system's social and technical impact.

CabPool | talanayush/CabPool (*Apr 2025*)

- **End-to-End App Development**: Built a campus-centric cab pooling platform with **ReactJS**, **Tailwind CSS**, **Node.js**, and **Express.js**.
- **Key Features**: Integrated in-app chat, automatic ticket expiry, JWT auth, and **UPI-based payment** with **PWA** support.
- **Database & API Design**: Designed and deployed scalable REST APIs using **MongoDB Atlas** for real-time ride management.

Atlas: Architecture Visualizer & Analyzer | Abhay1006/Reverse-Architect (*April 2026 – Present*)

- **Backend & Indexing**: Engineered a high-performance Express/TypeScript backend running a custom Language Server Protocol (LSP) pipeline and regex-based parsers to index and map codebases in 16+ languages.
- **Interactive Visualization**: Developed a dynamic 2D/3D WebGL canvas using React Flow, Three.js, and Zustand supporting animated directional particle flows, complexity heatmaps, and a syntax-highlighted code peek drawer.
- **Static Analysis**: Implemented graph-based heuristics including PageRank degree analysis and spreading activation to identify bug hotspots, code complexity patterns, and anti-patterns like circular dependencies.
- **Performance & Tooling**: Designed an SQLite caching layer with git-hash-based invalidation for instantaneous loading, and added exports for Markdown-compatible Mermaid flowcharts and JSON graph data.

Skills

- **Languages**: Proficient in C, C#, Python, C++, HTML5, CSS, JavaScript, TypeScript, and SQL.
- **Frameworks**: Experienced with **ReactJs**, **NextJs**, **NodeJs**, **Flask**, **Jinja2**, and **ExpressJs**.
- **Libraries**: Skilled in **TailwindCSS v3**,**Manim**,**11Labs**, **Mongoose**, and **Bootstrap v5**.
- **Tools/Utilities**: Proficient with Git, VS Code, MySQL, MongoDB, SQLite, Sqlalchemy, Linux, and Docker.

Achievements

- Earned a **3-star rating** (1744) on **CodeChef** by solving 900+ algorithmic challenges; profile available at CodeChef.
- Ranked as a **Knight** (1882) on **LeetCode** by solving 400+ problems, ranking in the top 5%; profile available at LeetCode.
- Completed 4000+ problems across various platforms, including Codolio; profile available at Codolio.

Certificates

- **SignLingo Research Paper**: Link.
- **100xdevs**: Web Development Course by Harkirat Singh Link.